THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The separation of cream from milk is done by: 1

(a) filtration **(b)** distillation **(c)** centrifugation **(d)** sublimation

2. A mixture of chalk powder in water can be separated by: 1

(a) chromatography **(b)** filtration **(c)** evaporation **(d)** sublimation

3. In a solution, the component present in the larger amount is usually the: 1

(a) precipitate **(b)** solute **(c)** colloid **(d)** solvent

4. The process used to separate a mixture of salt and ammonium chloride is: 1

(a) winnowing **(b)** decantation **(c)** sublimation **(d)** filtration

5. A solution of sugar can be separated into sugar and water by: 1

(a) evaporation (and condensation) **(b)** filtration
(c) magnet **(d)** winnowing

6. A solution that cannot dissolve more solute at a given temperature is called: 1

(a) colloidal **(b)** dilute **(c)** saturated **(d)** unsaturated

7. Which technique separates a mixture of two miscible liquids with different boiling points? 1

(a) filtration **(b)** distillation
(c) sedimentation **(d)** magnetic separation

8. Winnowing separates: 1

(a) iron from sand **(b)** husk from grain using wind
(c) gases **(d)** salt from water

9. The mass by mass percentage of a solution having 10 g of salt in 90 g of water is: 1

(a) 90 per cent **(b)** 11.1 per cent **(c)** 10 per cent **(d)** 9 per cent

10. Fractional distillation is used to separate: 1

(a) iron from sand **(b)** sand from water
(c) crude oil into petrol, kerosene, etc. **(d)** salt from sea water

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is distillation? Name the mixture it separates. 2

12. Give two examples of colloids and name their dispersed phase and medium. 2

13. What is the Tyndall effect? Give one example. 2

14. Why is the pencil line, not an ink line, used at the base in chromatography? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe fractional distillation of liquid air and name the gases collected in order. 3

16. Explain chromatography with an example of separating the dyes of black ink. 3

17. Why do wet clothes dry faster in the sun? Relate to evaporation of a solvent. 3

OR

Describe how a mixture of two immiscible liquids is separated with a separating funnel. 3

18. State the principle of each: 3
(a) filtration
(b) distillation
(c) sublimation.

OR

State the properties of alloys and give three examples with uses. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the process of distillation, simple and fractional, with applications and the apparatus. 5

OR

Explain how air is separated into its constituents and give the boiling points of nitrogen, argon and oxygen (approx). 5

20. Describe the separation methods for: 5
(a) sand and water
(b) cream and milk
(c) iron and sulphur
(d) salt and water
(e) camphor and sand.

OR

Discuss the importance of mixtures and their separation in daily life and industry with five examples. 5

END OF QUESTION PAPER — Total: 40 Marks

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